CHAPTER 95 — INSTRUMENT SYSTEM

CONTENTS — MAINTENANCE PROCEDURES

Paragraph Number	Title	Chapter/Section Number	Page Number
	INSTRUMENT SYSTEMS		
95-1	General	95-00-00	3
95-2	Panel mounted instruments	95-00-00	3
95-3	Removal/installation	95-00-00	3
95-4	Inspection/cleaning	95-00-00	3
95-5	Repair or replacement — instruments	95-00-00	3
95-6	Instrument panel	95-00-00	3
	FLIGHT INSTRUMENTS		
95-7	General	95-00-00	5
95-8	Electronic attitude director indicator (EADI)	95-00-00	5
95-9	Maintenance procedures	95-00-00	5
95-11	Pitot-static system	95-00-00	11
95-12	Pilot pitot-static system	95-00-00	11
95-13	Copilot pitot-static system	95-00-00	11
	NAVIGATION INSTRUMENTS		
95-14	General	95-00-00	13
95-15	Electronic horizontal situation indicator (EHSI)	95-00-00	13
95-16	Maintenance procedures	95-00-00	14
	MISCELLANEOUS INSTRUMENTS		
95-18	Digital clock	95-00-00	19
95-19	Functional Check	95-00-00	19
95-20	Operation	95-00-00	19
95-21	GMT (Greenwich Mean Time) Setting	95-00-00	19
95-22	LT (Local Time) Setting	95-00-00	19
95-23	ET (Elapsed Time) Count Up Setting	95-00-00	20
95-24	ET (Elapsed Time) Countdown Setting	95-00-00	20

FIGURES

Figure Number	Title	Page Number
95-1	Instrument panel	4
95-2	EADI displays and annunciators	6
95-3	EADI caution and failure annunciations (amber)	8
95-4	EADI failure annunciations (red)	9
95-5	Internal system failure (red)	10
95-6	Pitot-static system schematic	12
95-7	EHSI full compass displays and annunciators	15
95-8	EHSI full compass cautions and failure annunciations (amber)	17
95-9	EHSI full compass display annunciations (red)	18

INSTRUMENT SYSTEM

INSTRUMENT SYSTEMS

95-1. GENERAL.

Supplemental maintenance information for flight, navigation, propulsion, hydraulic, and miscellaneous instruments installed in search and rescue (SAR) configured model 412 helicopter is presented in the following sections.

95-2. PANEL MOUNTED INSTRUMENTS.

Refer to figure 95-1 for SAR instrument panel.

95-3. REMOVAL/INSTALLATION.

Refer to BHT-412-MM Chapter 95 for removal/installation of instrument panel mounted components.

95-4. INSPECTION/CLEANING.

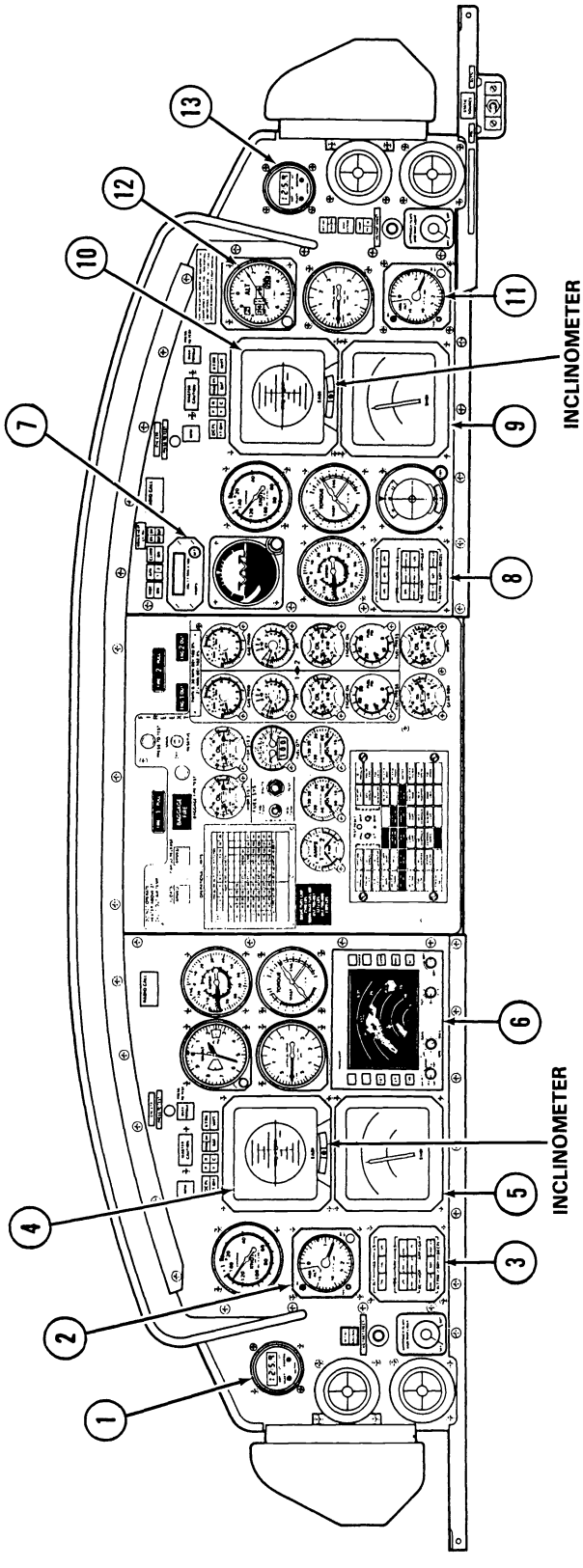
Refer to BHT-412-MM Chapter 95 for inspection/cleaning of instrument panel mounted components.

95-5. REPAIR OR REPLACEMENT — INSTRUMENTS.

Refer to BHT-412-MM Chapter 95 for repair or replacement of instrument panel mounted components.

95-6. INSTRUMENT PANEL.

Refer to BHT-412-MM Chapter 95 for removal/installation/adjustment of instrument panel mounted components.



1. Digital clock (copilot)
2. Radio altimeter indicator (copilot)
3. Mode selector (copilot)
4. EADI (copilot)
5. EHSI (copilot)
6. Color indicator (radar indicator)
7. Altitude preselect
8. Mode selector (pilot)
9. EHSI (pilot)
10. EADI (pilot)
11. Radar altimeter indicator (pilot)
12. Encoding altimeter
13. Digital clock (pilot)

Figure 95-1. Instrument panel

FLIGHT INSTRUMENTS

95-7. GENERAL.

Flight instruments that are unique to search and rescue configured helicopters include an electronic attitude director indicator.

95-8. ELECTRONIC ATTITUDE DIRECTOR INDICATOR (EADI).

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EADI provides the traditional sphere-type attitude display with lateral and vertical steering signals to assist the pilot in intercepting and maintaining a desired flightpath.

The EADI (figure 95-2) displays the following information:

- Attitude
- Pitch and Roll
- Command Bars
- Collective Command Cues
- Glideslope Deviation
- Expanded Localizer Deviation
- Marker Beacon Annunciator
- Radio Altitude
- — Rising Runway
- — Digital Readout
- Decision Height Setting/Warning
- Flight Director and SAR Mode Annunciators
- Rate-of-Turn

In addition to the familiar ADI information the EADI displays the following amber caution and failure annunciations (figure 95-3):

- Attitude Source Annunciator
- Flight Director Failure Annunciator
- Decision Height Warning Annunciator
- Flight Management System Message Annunciator
- Radio Altitude Failure Annunciator
- Comparision Monitor Annunciator
- Decision Height Failure Annunciator
- Common Symbol Generator Annunciator

The EADI also displays the following red failure annunciations (figure 95-4):

- Attitude Failure
- Glideslope, Glidepath or VNAV, Expanded Localizer, or Rate-of-Turn
- Internal Failure

1. Failure of the attitude source results in the pitch scale and roll pointer being removed, the sphere will be displayed blue, and a red ATT FAIL annunciator being displayed.

2. Failure of the glideslope, glidepath or VNAV, expanded localizer or azimuth, or rate-of-turn will result in the pointer being removed and a red X drawn through the scale. The letter G or V at the zero deviation position identifies the invalid information.

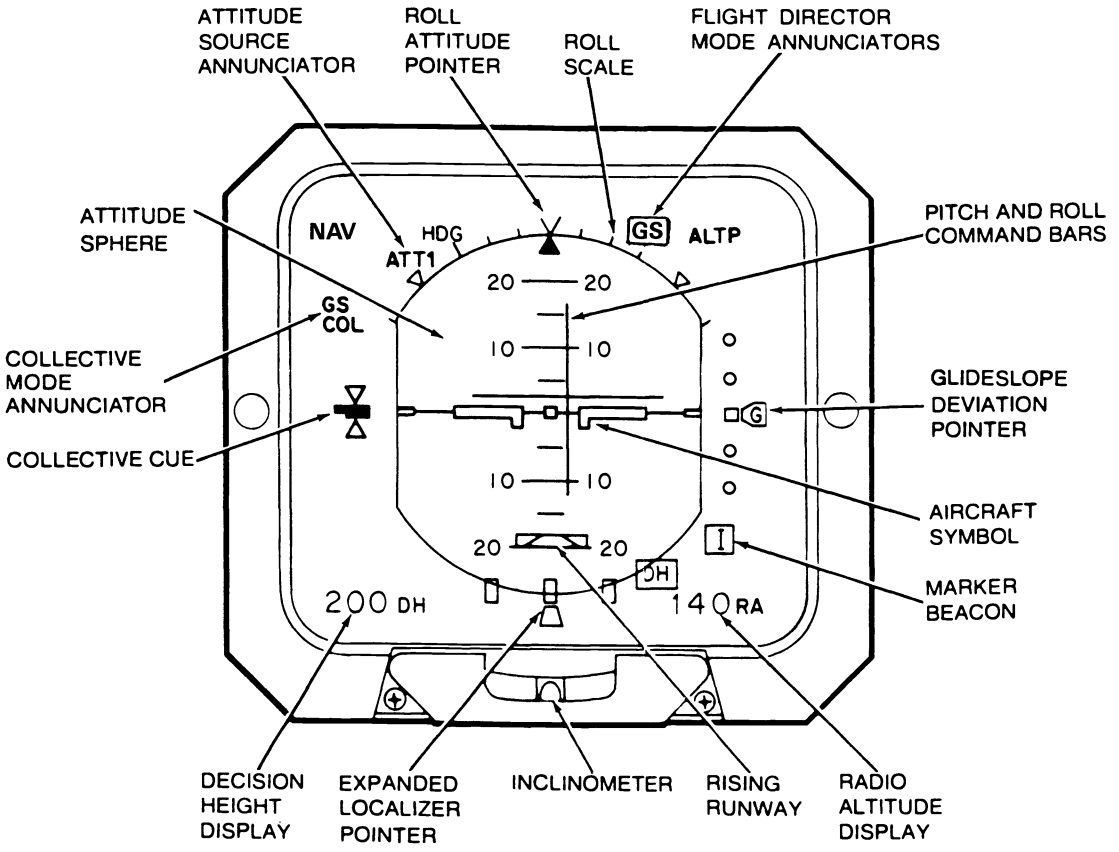
3. An extended failure of the symbol generator will cause the EADI to go blank. A failure of the input/processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on component displays/annunciation and operation of the EADI, refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance, Manual Publication Number A09-1169-024.

95-9. MAINTENANCE PROCEDURES.

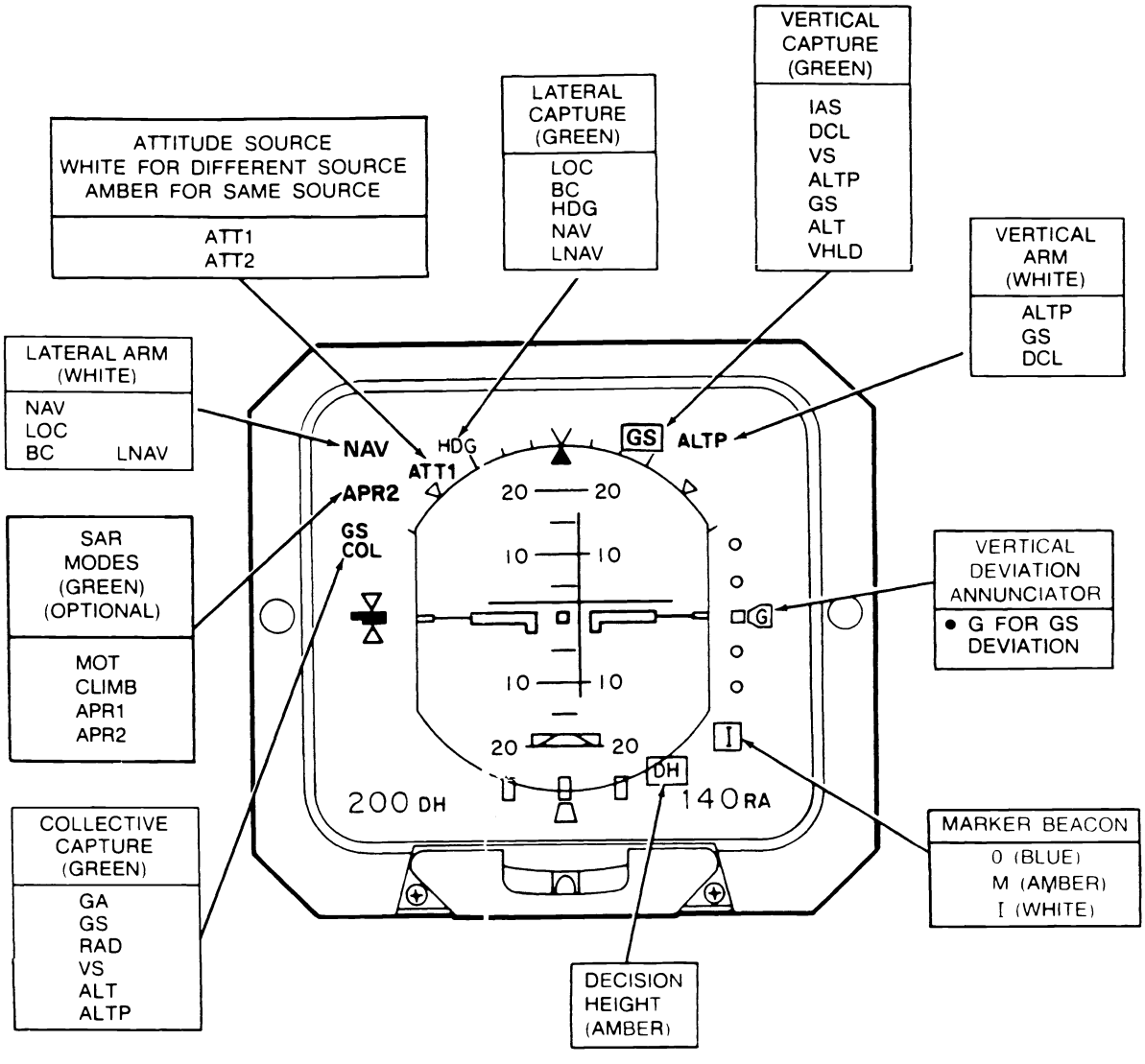
95-10. EADI.

Refer to BHT-412-MM Chapter 95 for maintenance procedures of panel mounted instrument.



412-SAR-95-2-1

Figure 95-2. EADI displays and annunciators (Sheet 1 of 2)



412-SAR-95-2-2

Figure 95-2. EADI displays and annunciators (Sheet 2)

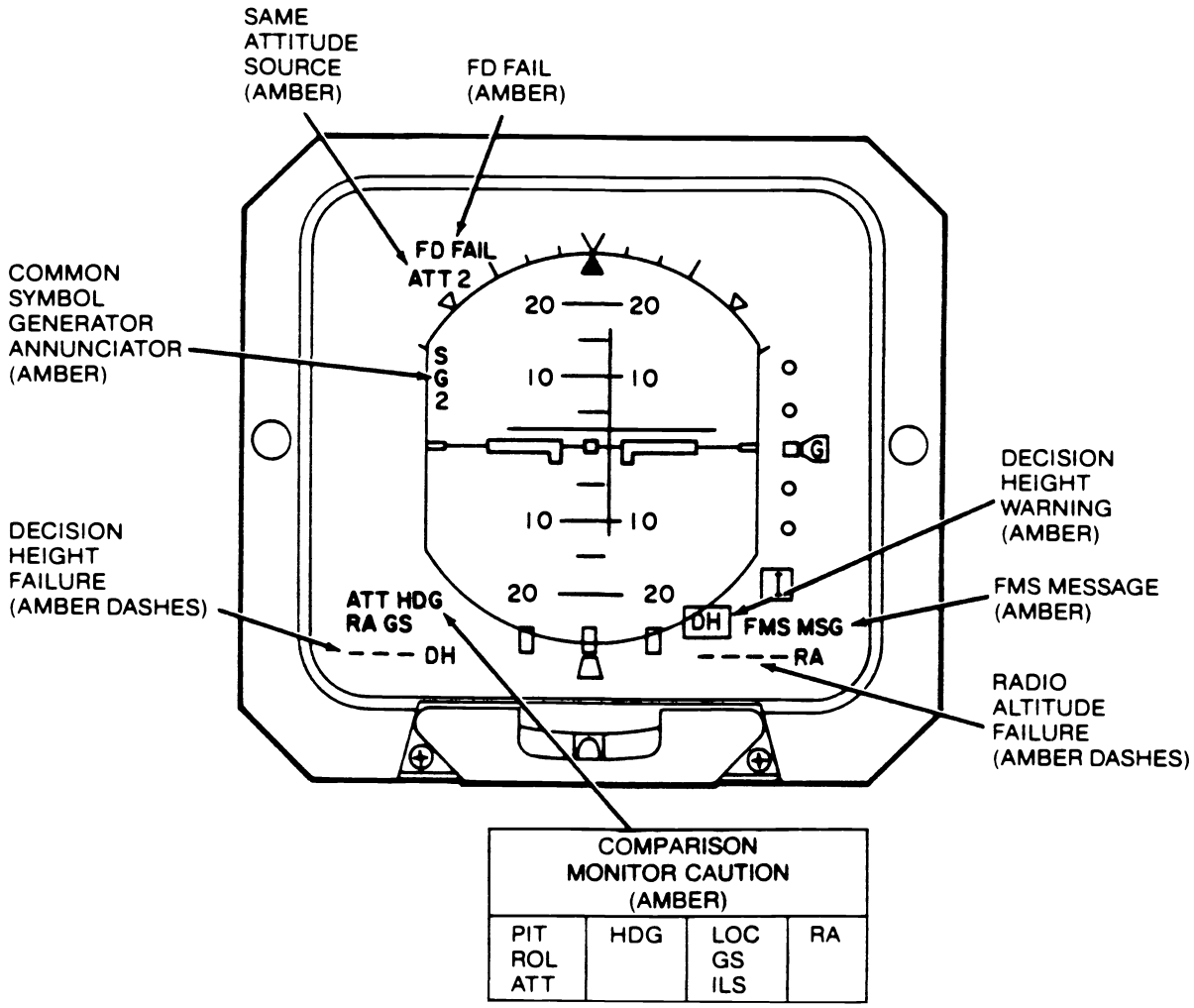
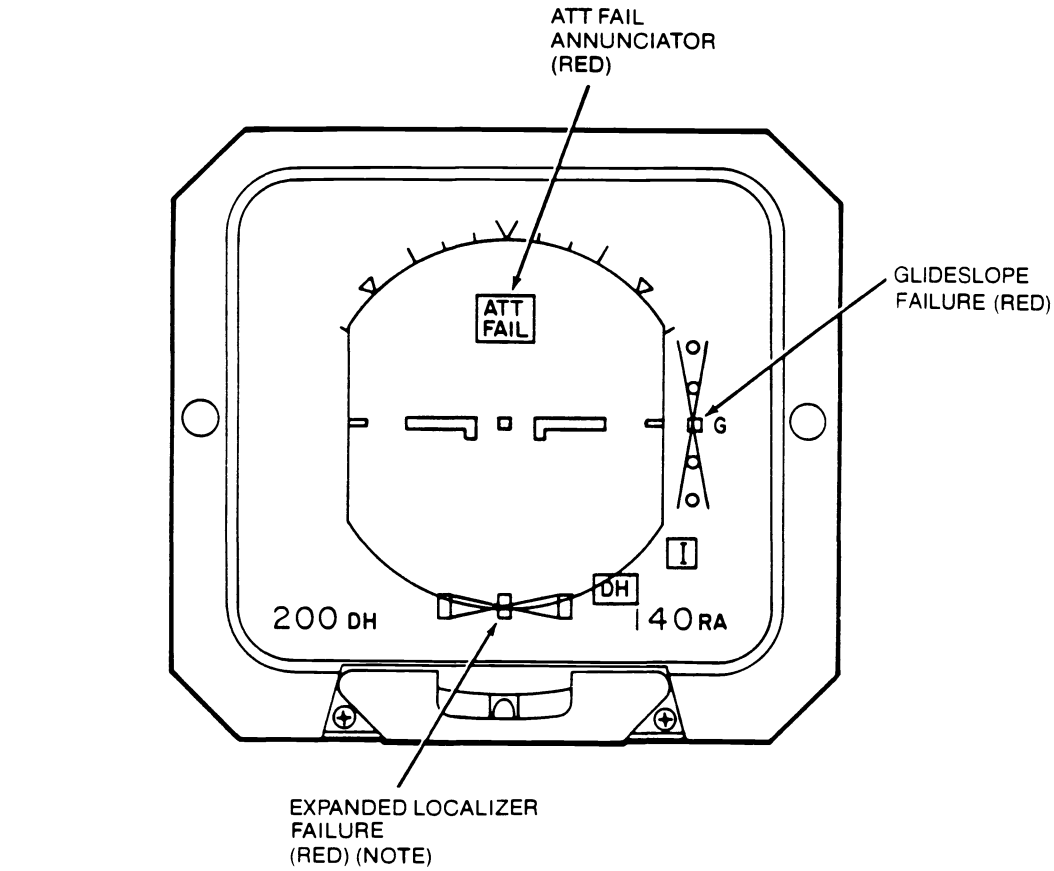


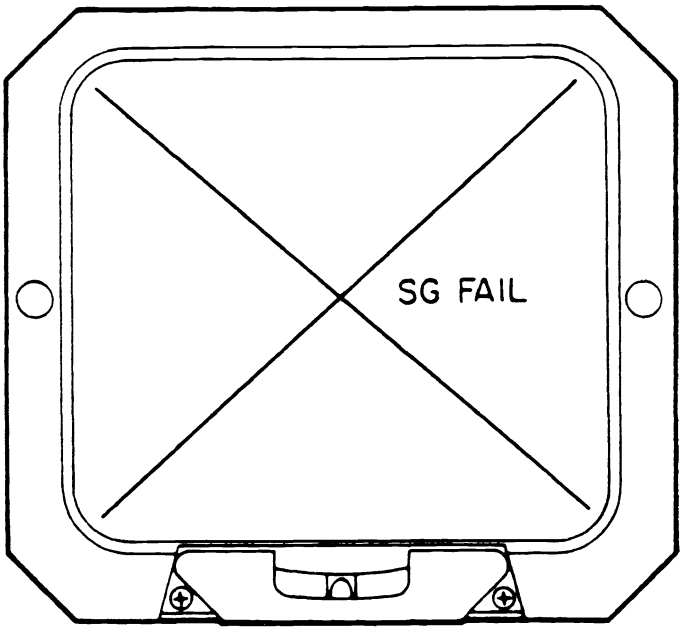
Figure 95-3. EADI caution and failure annunciations (amber)



NOTE: RATE-OF-TURN FAILURE IS SIMILAR.

412-SAR-95-4

Figure 95-4. EADI failure annunciations (red)



NOTE: COMPLETE FAILURE OF A SYMBOL GENERATOR WILL RESULT IN BLANK EADI AND EHSI DISPLAYS.

412-SAR-95-5

Figure 95-5. Internal system failure (red)

95-11. PITOT-STATIC SYSTEM.

95-12. PILOT PITOT-STATIC SYSTEM.

1. The pilot pitot-static system (figures 95-6) consists of right pitot tube, right and left static ports, right roof-mounted static port, alternate static air source, pilot airspeed indicator, barometric encoding altimeter, vertical speed indicator, air data sensor, and appropriate tubing and hardware for interconnection of system components.

2. The right pitot tube supplies pitot pressure (impact air) directly to pilot airspeed indicator and air data sensor. Upper right, lower left and right roof-mounted static ports are connected together and supply equalized static pressure to pilot airspeed indicator, barometric encoding altimeter, vertical speed indicator and air data sensor. Pitot tube and static ports are equipped with electric heaters to provide all-weather capability.

3. An alternate static air source is provided by a pneumatic valve installed below the pilot instrument panel. Tubing and fittings are included to connect pneumatic valve to pilot static ports and instruments. With the toggle of the pneumatic valve set to PRI, static air is supplied to instruments from outside pilot static air ports. With the toggle set to ALTN, an alternate static air source (inside cabin air) is selected and outside static ports are shut off.

95-13. COPILOT PITOT-STATIC SYSTEM.

1. The copilot pitot-static system (figures 95-6) consists of left pitot tube, right and left static ports, left roof-mounted static port, copilot airspeed indicator, barometric altimeter, vertical airspeed indicator, air data sensor, air data computer, and appropriate tubing and hardware for interconnection of system components.

2. The left pitot tube supplies pitot pressure (impact air) directly to copilot airspeed indicator, air data sensor, and air data computer. Lower right, upper left and left roof-mounted static ports are connected together and supply equalized static pressure to copilot airspeed indicator, barometric altimeter, vertical speed indicator, air data sensor, and air data computer. Pitot tube and static ports are equipped with electric heaters to provide all-weather capability.

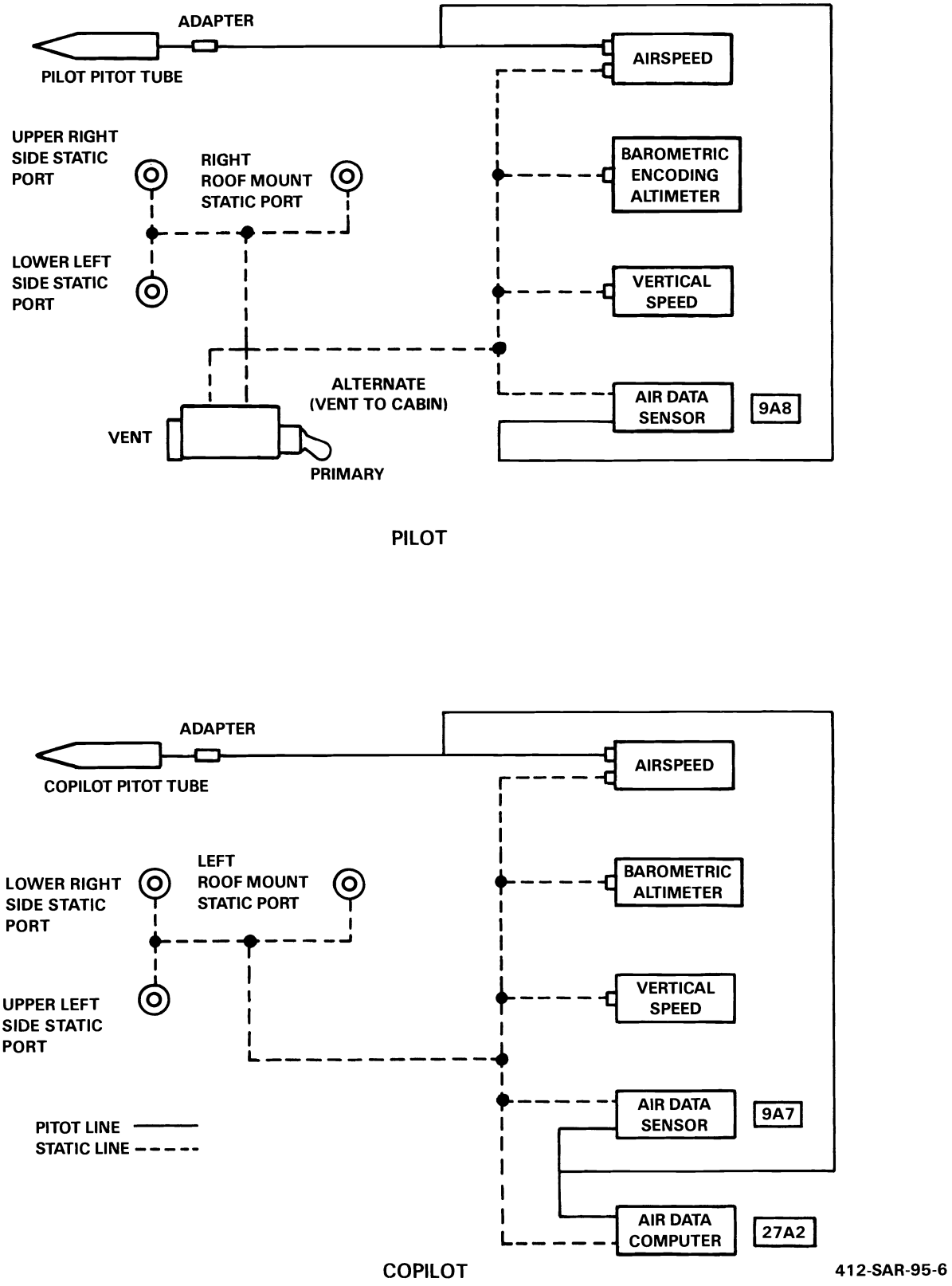


Figure 95-6. Pitot-static system schematic

NAVIGATION INSTRUMENTS

95-14. GENERAL.

Navigation instruments are electrically powered, except for the standby compass.

95-15. ELECTRONIC HORIZONTAL SITUATION INDICATOR (EHSI).

Refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024 for component description and operation.

The EHSI provides compass displays and a map-like display of the helicopter position relative to VOR radials, localizer, glideslope beam, or FMS waypoints.

The EHSI (figure 95-7) provides the following displays and annunciators:

- Course select/desired
- Drift angle bug
- FMS approach message
- Heading source annunciator
- Heading select bug and heading select display
- Navigation source annunciators
- Distance display
- Preselected Nav source annunciator
- Course preselect pointer
- Compass card and fore and aft lubber lines
- Glideslope deviation pointer
- Groundspeed, time-to-go, or elasped time display
- Bearing pointers and source annunciators
- Reciprocal course pointer
- Aircraft symbol
- Course deviation bar
- Weather radar mode and tilt annunciator
- To-From arrow
- Course select pointer
- Compass sync annunciator.

In addition to the standard displays and annunicators, the EHSI also displays the following amber caution and failure annunciations (figure 95-8):

- Waypoint alert annunciator
- Same heading or navigation source
- DME hold annunciator
- Weather radar (WX) failure
- Displays failures
  — amber dashes replace the digital display for the following systems when they fail.
- — Groundspeed (GSPD)
- — Time-To-Go (TTG)
- — Elapsed Time (ET)
- — Distance Measuring Equipment (DME)
- — Heading Select (HDG)
- — Course Select (CRS)
- — Desired Track (DTRK).

The EHSI also displays the following red failure annunciations (figure 95-9):

- Heading failure
- Course deviation
- Glideslope deviation.

1. Failure of the heading system is indicated by the removal of the bearing pointers, TO/FROM arrow, select course pointer, course deviation bar, and course scale. The digital select course and digital heading select readouts are dashed and a red HDG FAIL is displayed. The heading source annunciator changes to a white HDG 1 through HDG 4 to include which heading source has failed.

2. A failure of the course deviation, or glideslope deviation results in the removal of the deviation pointer, and a red X drawn through the scale.

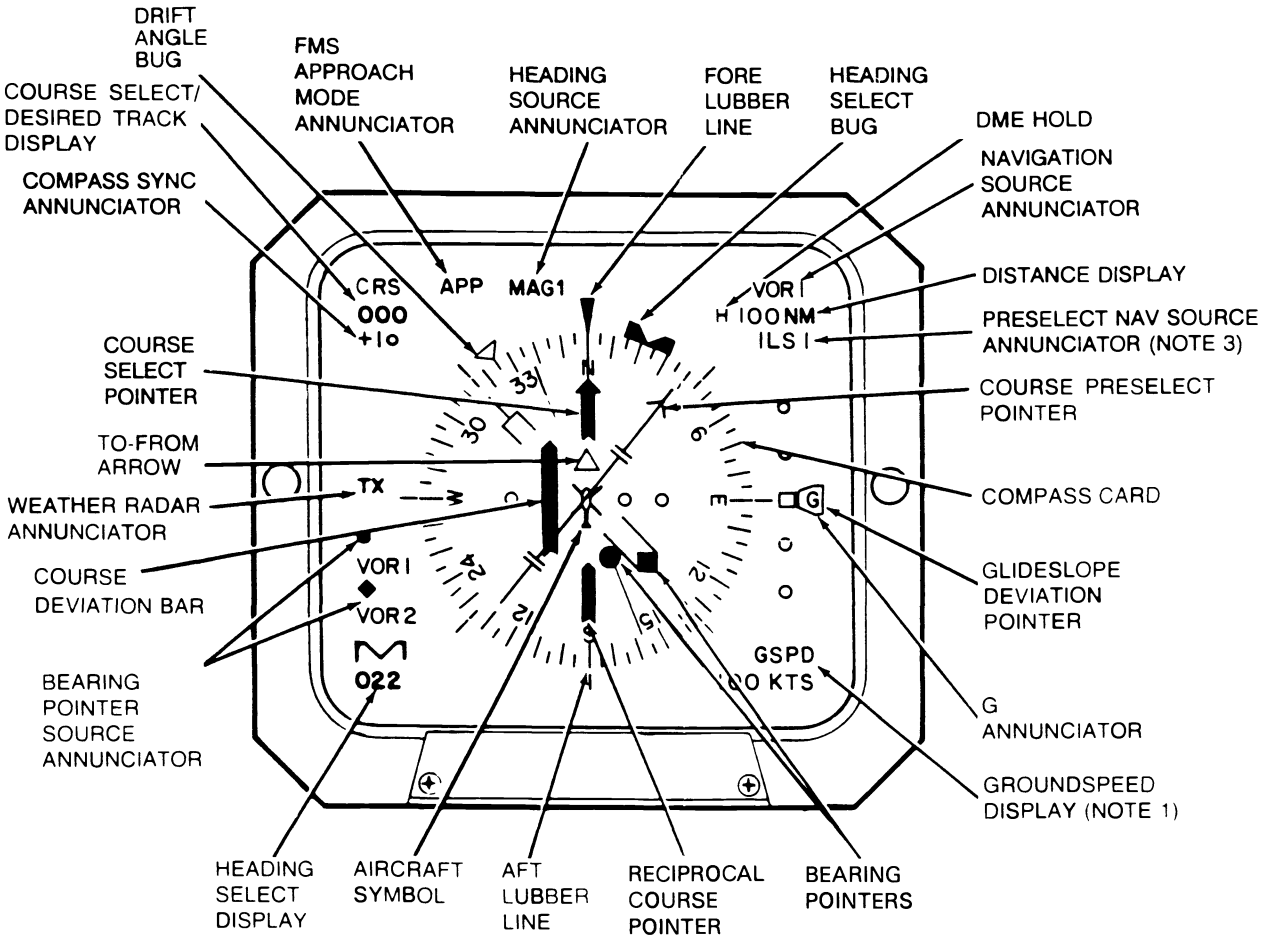
3. An internal failure of the symbol generator will cause the EHSI to go blank. A failure of the input/processor results in a red X centered on the display, with annunciation SG FAIL.

For additional information on EHSI displays/annunciations and operation refer to Honeywell, SPZ-7600 Integrated Flight Control System Bell 412HP System Maintenance Manual, Publication Number A09-1169-024.

95-16. MAINTENANCE PROCEDURES.

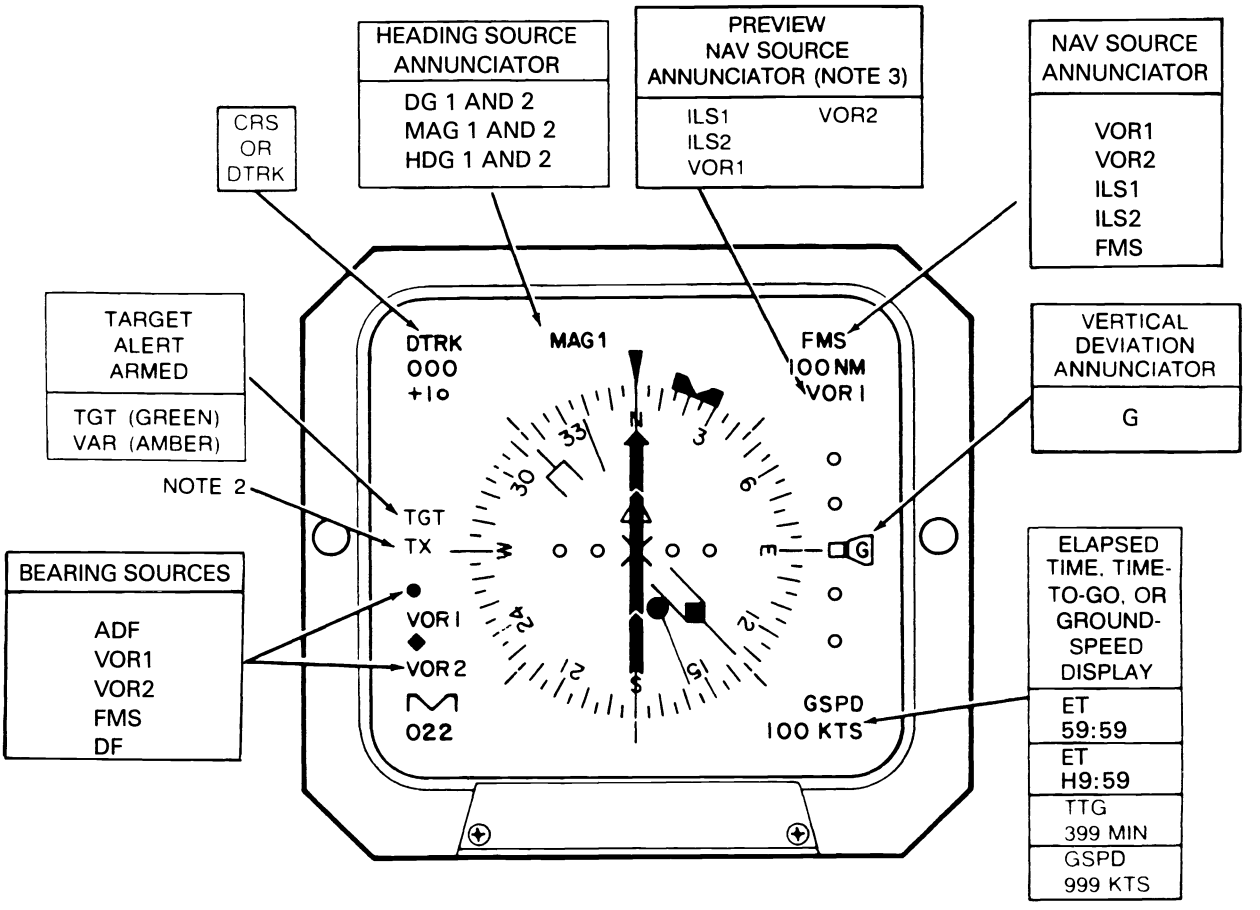
95-17. EHSI.

Refer to BHT-412-MM Chapter 95 for maintenance procedures of panel mounted instrument.



412-SAR-95-7-1

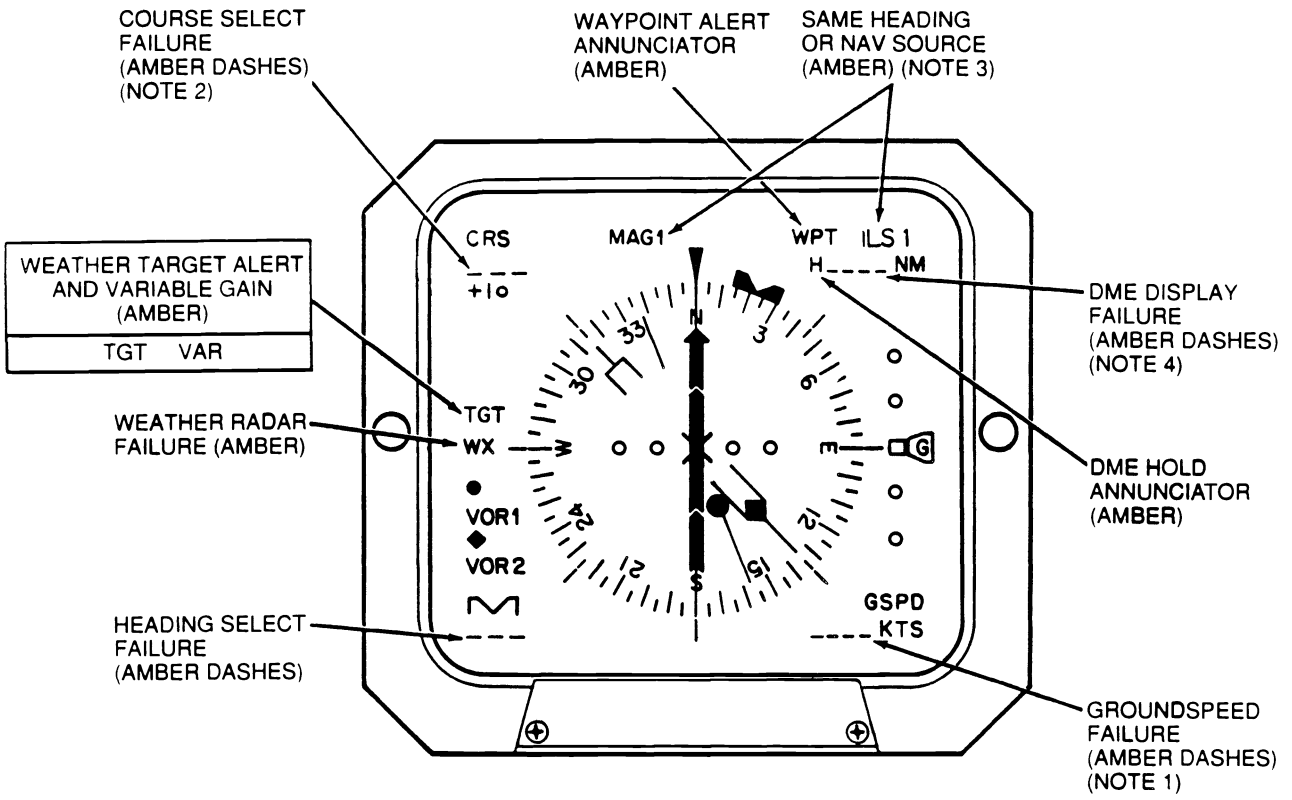
Figure 95-7. EHSI full compass displays and annunciators (Sheet 1 of 2)



NOTES:
1. TIME-TO-GO AND ELAPSED TIME ARE ALSO DISPLAYED AT THIS LOCATION.
2. TX APPEARS WHENEVER THE WEATHER RADAR IS TRANSMITTING AND WX IS NOT SELECTED FOR DISPLAY.
3. DME IDENT DISPLAYED WHEN SELECTED AS NAV SOURCE.

412-SAR-95-7-2

Figure 95-7. EHSI full compass displays and annunciators (Sheet 2)

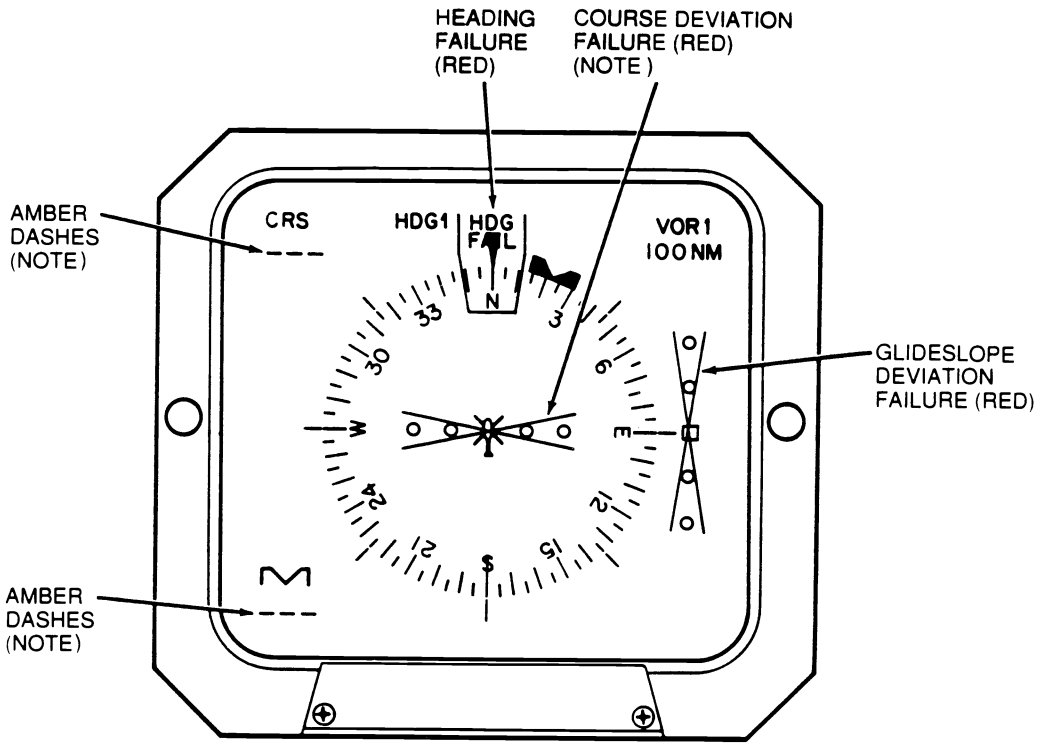


NOTES:

1. TTG AND ET FAILURES ARE SIMILAR
2. DTRK FAILURE IS SIMILAR.
3. IF BOTH THE PILOT AND THE COPILOT SELECT THE CROSS-SIDE NAV SOURCE, THE NAV ANNUNCIATORS ARE AMBER.
4. AN AMBER N/A IS DISPLAYED WHEN SELECTED LRN DISTANCE IS NOT AVAILABLE.
5. THIS DIAGRAM IS FOR ILLUSTRATION PURPOSES ONLY. IT DOES NOT NECESSARILY REPRESENT AN ACTUAL PRESENTATION.

412-SAR-95-8

Figure 95-8. EHSI full compass cautions and failure annunciations (amber)



NOTE

IN THE EVENT OF A HEADING FAILURE, THE COURSE SCALE AND RED X WILL NOT BE DISPLAYED AND THE CRS AND HDG READOUTS WILL INDICATE AMBER DASHES.

412-SAR-95-9

Figure 95-9. EHSI full compass display annunciations (red)

MISCELLANEOUS INSTRUMENTS

95-18. DIGITAL CLOCK.

The clock has a liquid crystal display, lighting system, and ultraviolet filter to provide uniform night illumination and protection from sunlight. The clocks two push buttons control the four displays

- GMT = Greenwich Mean Time (24 hour format)
- LT = Local Time (12 or 24 hour format)
- ET = Elapsed Time Counter (1 second to 99 hours and 59 minutes range)
- ET = Elapsed Time Countdown Timer (1 second to 1 hour range)

95-19. FUNCTIONAL CHECK.

Press and hold down SELECT button for 3 seconds. The display will read '88.88' and GMT, LT, and ET will be illuminated.

95-20. OPERATION.

1. SELECT Button. The SELECT button sequentially pressed will illuminate GMT (Greenwich mean time), LT (local time), and ET (elapsed time).

2. CONTROL Button. The CONTROL button sequentially pressed (after pressing the SELECT and CONTROL button simultaneously) will change the display. Pressing the button momentarily will start and reset ET (elapsed time).

95-21. GMT (GREENWICH MEAN TIME) SETTING.

1. Press SELECT buttom momentarily until GMT is illuminated.

2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).

3. Press CONTROL button once to display first digit to be set.

4. Press CONTROL button momentarily to set digit. Each button push advances the digit.

5. Press SELECT button once to select the next digit to be set.

6. Perform steps 3, 4, and 5 above until remaining digits are set.

7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

95-22. LT (LOCAL TIME) SETTING.

1. Press SELECT buttom momentarily until LT is illuminated.

2. Press SELECT and CONTROL buttons once simultaneously to enter set mode (tens of hours digit starts flashing).

3. Press CONTROL button once to display first digit to be set.

4. Press CONTROL button momentarily to set digit. Each button push advances the digit.

5. Press SELECT button once to select the next digit to be set.

NOTE

Minutes are synchronized with GMT and can not be changed in the LT (local time) setting mode.

If minutes require changing follow GMT procedures for minutes.

6. Perform steps 3, 4, and 5 above until remaining digits are set.

7. Press SELECT button once to exit the set mode (flashing display indicates clock is running).

95-23. ET (ELAPSED TIME) COUNT UP SETTING.

1. Press SELECT buttom momentarily until ET is illuminated.

2. Press CONTROL button once to start count up. Counter will start counting in minutes and seconds up to 59 minutes and 59 seconds. Then the counter will start counting in hours and minutes up to 99 hours and 59 minutes.

3. Press CONTROL button once to reset the ET count up to zero.

95-24. ET (ELAPSED TIME) COUNTDOWN SETTING.

1. Press SELECT buttom momentarily until ET is illuminated.

2. Press SELECT and CONTROL buttons once simultaneously to enter set mode.

3. Press CONTROL button once to display first digit to be set.

4. Press CONTROL button momentarily to set digit. Each button push advances the digit up from 1 second to 59 minutes and 59 seconds.

5. Press SELECT button once to select the next digit to be set.

6. Perform steps 3, 4, and 5 above until remaining digits are set.

7. Press SELECT button once to exit the set mode.

8. Press CONTROL button once to start the countdown. Display will flash and an external alarm will be activated when the counter reaches zero.

9. Press SELECT or CONTROL button to deactivate the flashing display and external alarm.